Persuading an inattentive and privately informed receiver

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ASSET

Inattention and persuasion

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This paper studies the persuasion of a **receiver** who is privately informed about costs and benefits of information.

Overview

This paper studies persuasion à la Kamenica and Gentzkow when Receiver is **privately informed** and **rationally inattentive**.

Do screening devices help Sender? Which types of Receiver pay attention?

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Results:

1. Persuasion mechanisms are equivalent to experiments;
2. Upper censorships are optimal in applications;
3. Persuasion mechanisms are without loss under *deterministic* coordination mechanisms,
   but with loss under *stochastic* coordination mechanisms.

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Plan:

1. Model;
2. Literature;
3. Equivalence between persuasion mechanisms and experiments;
4. Optimal experiment with single-peaked outside option;
5. Generalized principal-agent setting (Myerson, 1982).

Model

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Sender and Receiver.

Payoff to Receiver from **action** $a \in \{0, 1\}$ and **effort** $e \in [0, 1]$ is

$$U_R(a, \theta, e, c, \lambda) = \underbrace{a(\theta - c)}_{\text{material payoff } u_R} - \underbrace{\lambda k(e)}_{\text{effort cost}},$$

in which:

- $\theta \in [0, 1]$ is the state, drawn from a continuous F_0, with mean x_0;
- $c \in [0, 1]$ is the outside option;
- $\lambda \geq 0$ is the attention cost;
- $k : [0, 1] \to \mathbb{R}_+$ is continuous.

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Payoff to Sender from action a is $U_S(a) = a$.

Receiver is privately informed about her **type**: (c, λ), drawn from H with support T, independently of θ.

Persuasion mechanisms

A signal is $\sigma : \Theta \to \Delta M$.

A persuasion mechanism is a list of signals $(\sigma_r)_{r \in T}$.

Timing

1. Sender commits to a persuasion mechanism $(\sigma_r)_{r \in T}$;
2. Receiver chooses a signal σ_r and effort e, knowing her type;
3. Nature draws θ according to F_0;
   With probability e, Receiver observes a message m drawn from $\sigma_r(\theta)$;
4. Receiver chooses an action a.

Literature

Bayesian persuasion

Kamenica and Gentzkow (2011)... **Kolotilin et al. (2017)**, Kolotilin (2018), Guo and Shmaya (2019), Kolotilin et al. (2022).

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Persuasion with "outside information"

Bizzotto et al. (2020); Dworczak and Pavan (2022); Matysková and Montes (2023).

Dynamic information provision

Liao (2021); Che et al. (2023); Au and Whitmeyer (2023); Knoepfle (2024); Hébert and Zhong (2025); Jain and Whitmeyer (2025)...

Attention management

Lipnowski et al. (2020).

Equivalence between persuasion mechanisms and signals

Mean distribution

Fix a signal.

Optimal action given belief μ of type-(c, λ) Receiver is:

$$a^\star(\mu) \in \left[\mathbf{1}\{\mathbb{E}_{\theta\sim\mu}[\theta] > c\}, \mathbf{1}\{\mathbb{E}_{\theta\sim\mu}[\theta] \geq c\}\right].$$

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$$B(c) = \underbrace{\mathbb{E}_{\mu\sim p}\left[\mathbb{E}_{\theta\sim\mu}[u_R(a^\star(\mu), c, \theta)]\right]}_{\text{payoff of informed action}} - \underbrace{u_R(a^\star(F_0), c, x_0)}_{\text{payoff of uninformed action}}.$$

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The mean distribution describes equilibrium action and effort.

Signals as information policies

We identify a signal with its **information policy**.

The *information policy* of the CDF F is the function $I_F : \mathbb{R}_+ \to \mathbb{R}_+$ given by

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(1) I_F is convex;

(2) $I_{\overline{F}} \leq I_F \leq I_{F_0}$, in which:

- $\overline{F}$ is the CDF induced by an **uninformative signal**.
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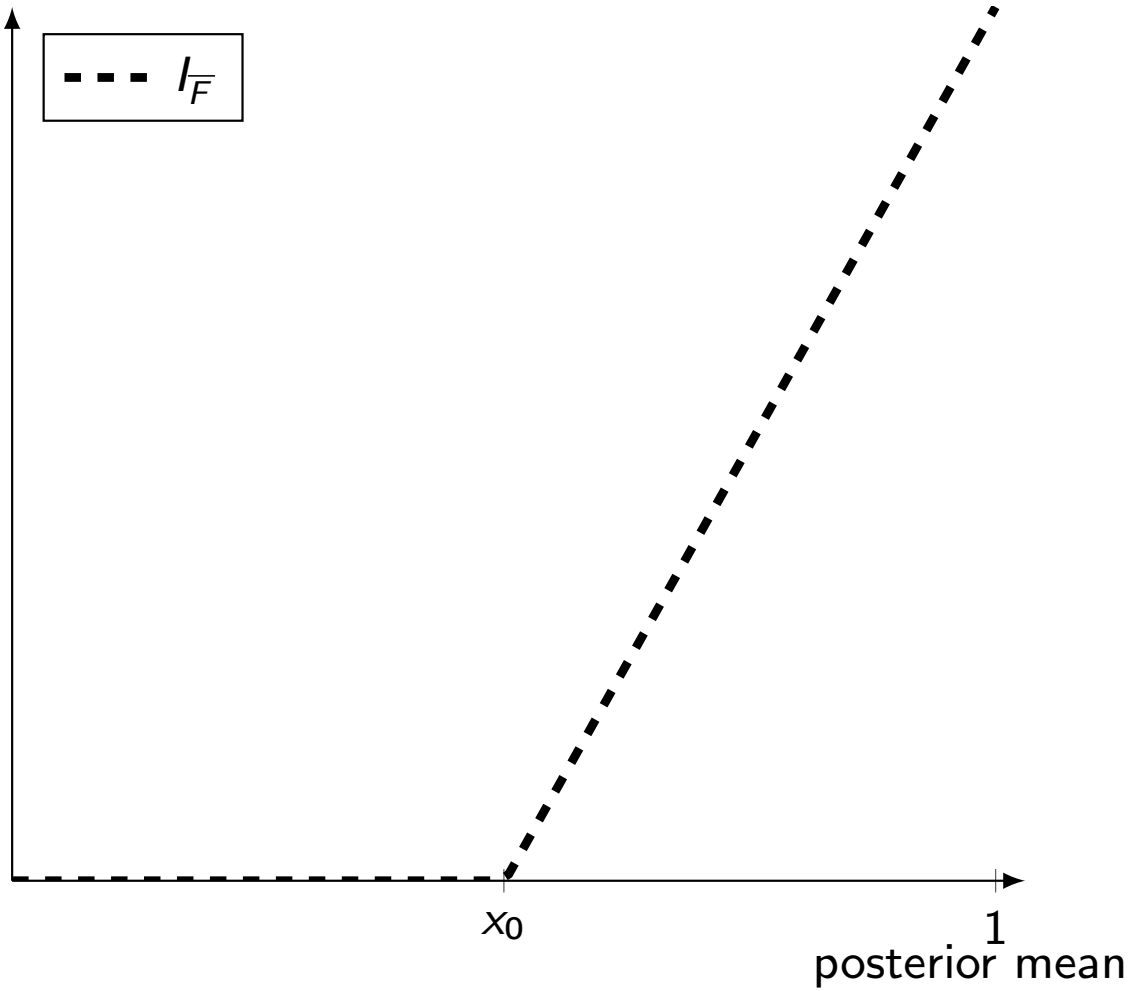
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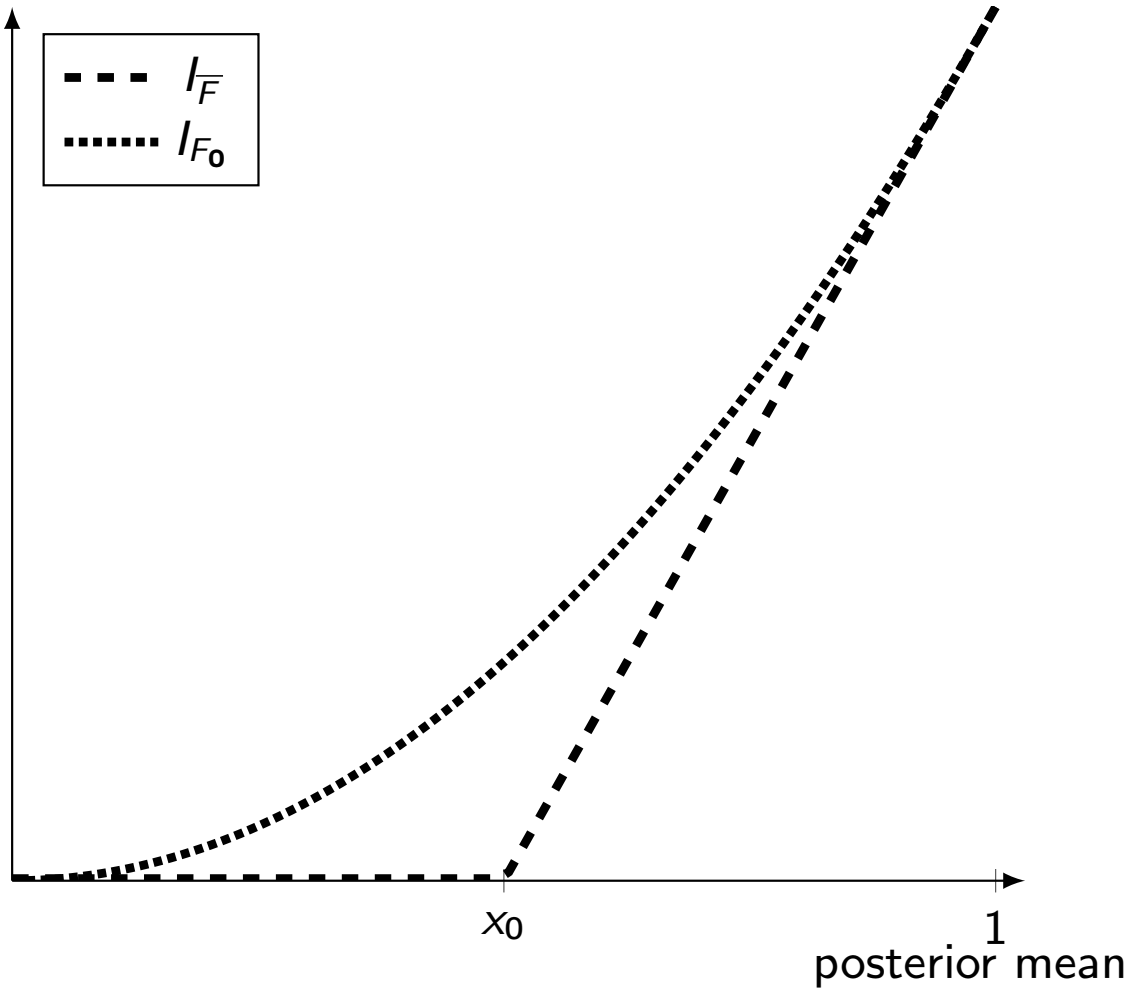
Lemma 1 (Gentzkow and Kamenica, 2016; Kolotilin, 2018.)

If $I : \mathbb{R}_+ \to \mathbb{R}_+$ satisfies (1) and (2), then there exists a signal that induces a posterior-mean CDF with information policy I.

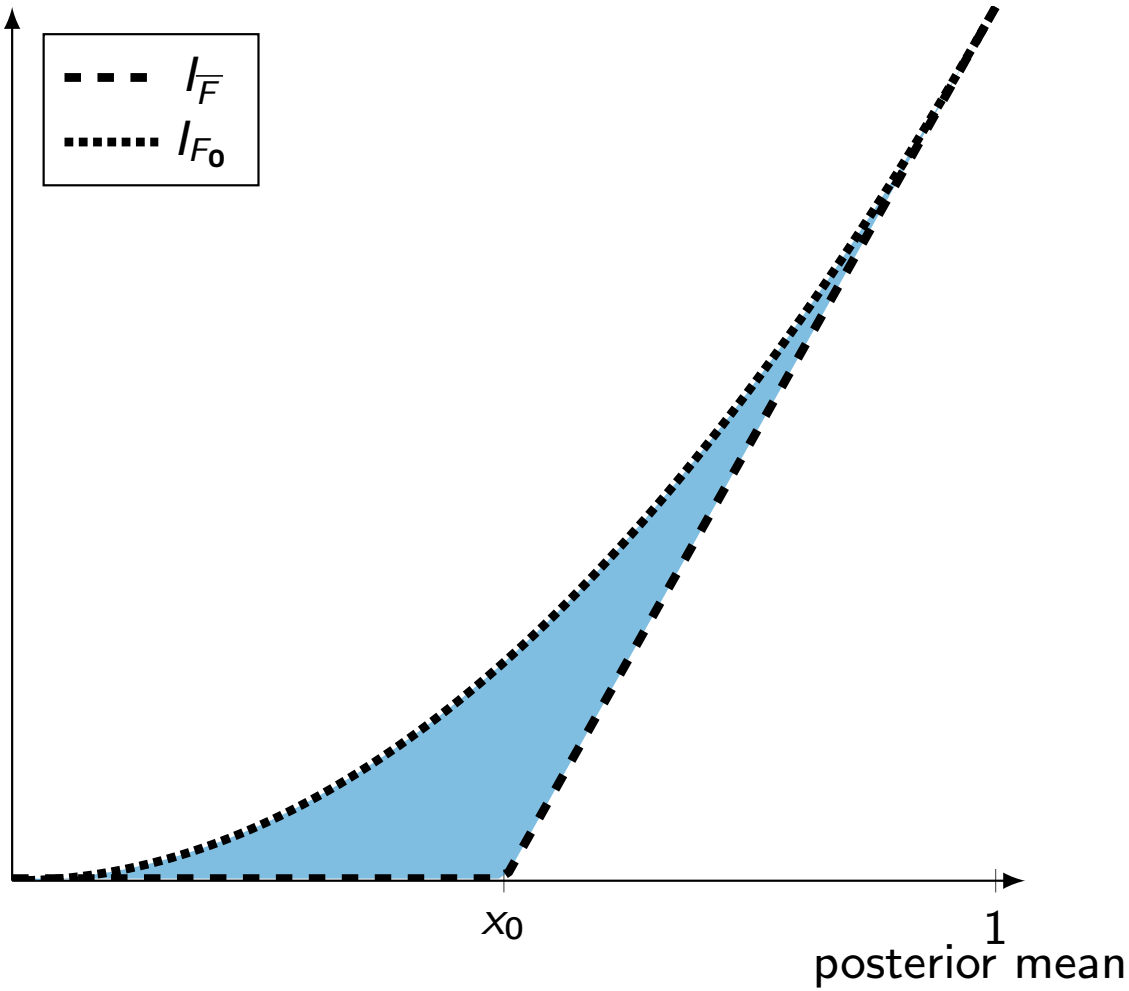
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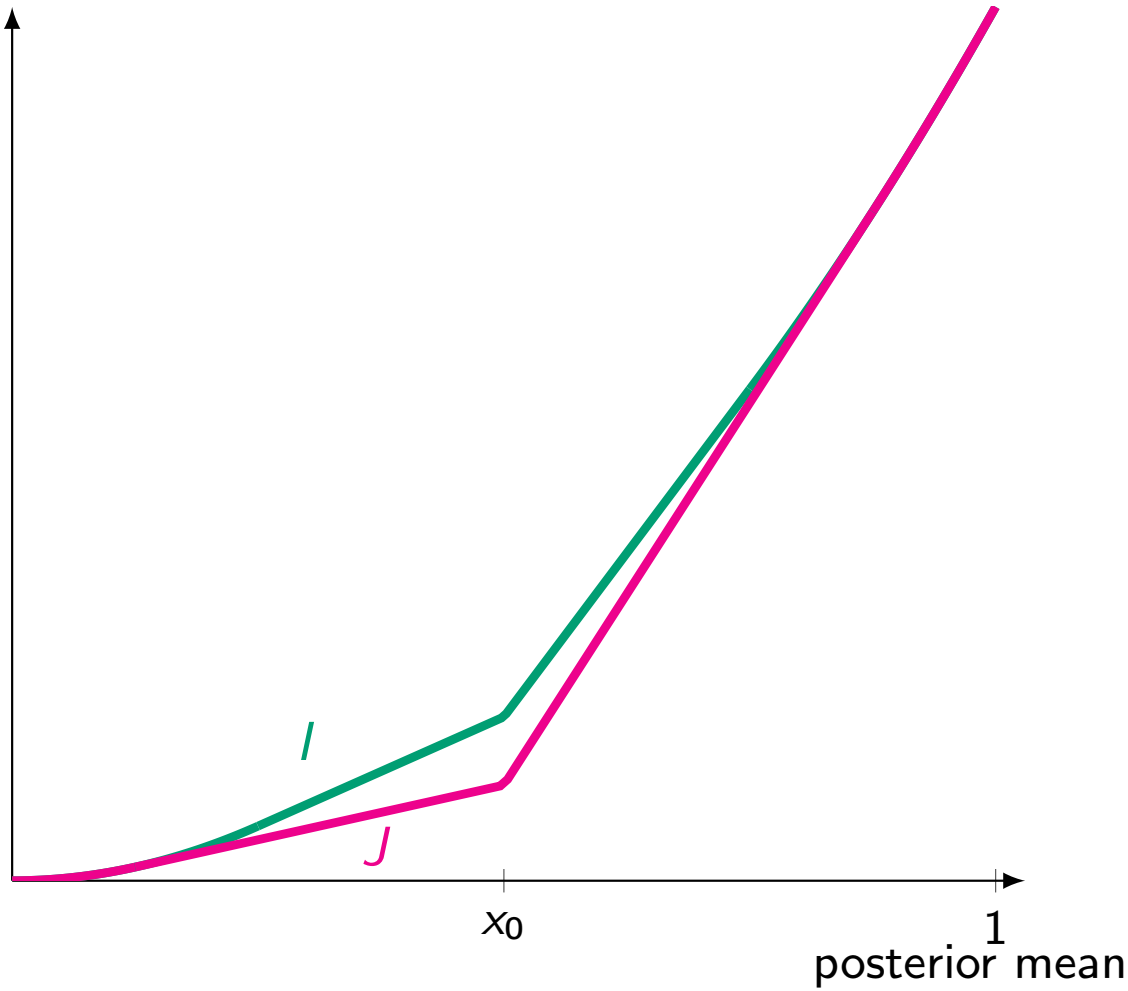
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Information policies



I is "more informative" than J if: $I \geq J$.

Interval structure of the extensive margin

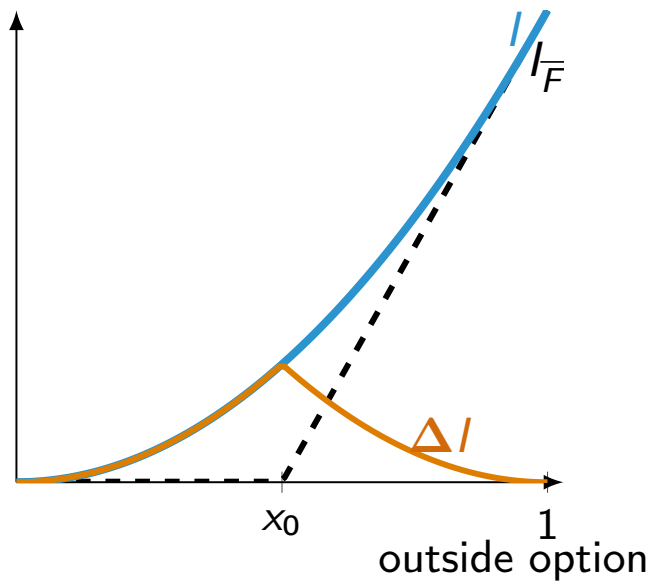
The *net informativeness* of I is denoted by

$$\Delta I = I - I_{\overline{F}}.$$

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The marginal benefit of effort given information policy I satisfies, for all c,

$$B(c) = \Delta I(c).$$



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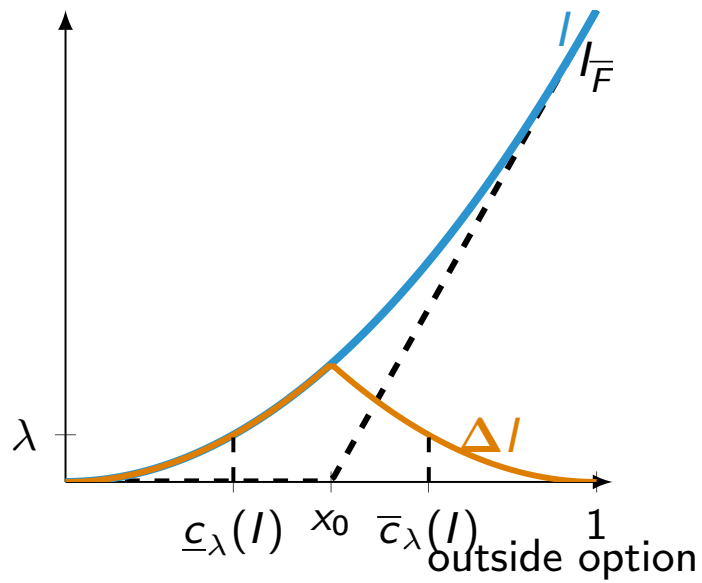
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If **cost is linear** ($k(e) = e$), then there are *cutoff types* $\underline{c}_\lambda(I)$, $\overline{c}_\lambda(I)$ such that intermediate outside options are attentive.

Persuasion mechanisms

The *value of information policy* I is the interim payoff of type-(c, λ) Receiver:

$$V_R(\boldsymbol{\Delta} I(c), \lambda) := \max_{e \in [0,1]} \underbrace{e \boldsymbol{\Delta} I(c)}_{\substack{\text{strictly} \\ \text{increasing} \\ \text{differences}}} - \lambda k(e).$$

- Information and attention are **complements**.

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A *persuasion mechanism* is a menu of information policies: $(I_r)_{r \in T}$.

A persuasion mechanism $(I_r)_{r \in T}$ is *incentive compatible* (IC) if:

for all types $(c, \lambda) \in T$ and reports $\widetilde{(c, \lambda)} \in T$, we have

$$V_R(\boldsymbol{\Delta} I_{(c,\lambda)}(c), \lambda) \geq V_R(\boldsymbol{\Delta} I_{\widetilde{(c,\lambda)}}(c), \lambda).$$

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An IC persuasion mechanism $(I_r)_{r \in T}$ is *equivalent to information policy* J if, for every type (c, λ), the optimal effort and action are the same:

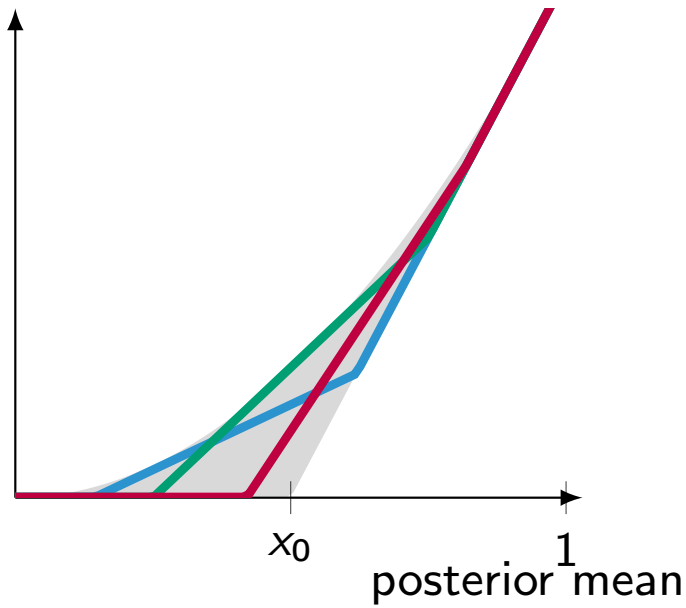
1. $\operatorname*{argmax}_{e \in [0,1]} e \mathbf{\Delta} I_{(c,\lambda)}(c) - \lambda k(e) \subseteq \operatorname*{argmax}_{e \in [0,1]} e \mathbf{\Delta} J(c) - \lambda k(e)$,
2. if $(0, 1] \cap \operatorname*{argmax}_{e \in [0,1]} e \mathbf{\Delta} I_{(c,\lambda)}(c) - \lambda k(e) \neq \varnothing$, then $\partial I_{(c,\lambda)}(c) \subseteq \underbrace{\partial J(c)}_{\substack{[J'_-(c), J'_+(c)] \\ \text{subdifferential}}}$.

Equivalence

Proposition 1

For every IC persuasion mechanism, there exists an equivalent information policy.

Proof sketch



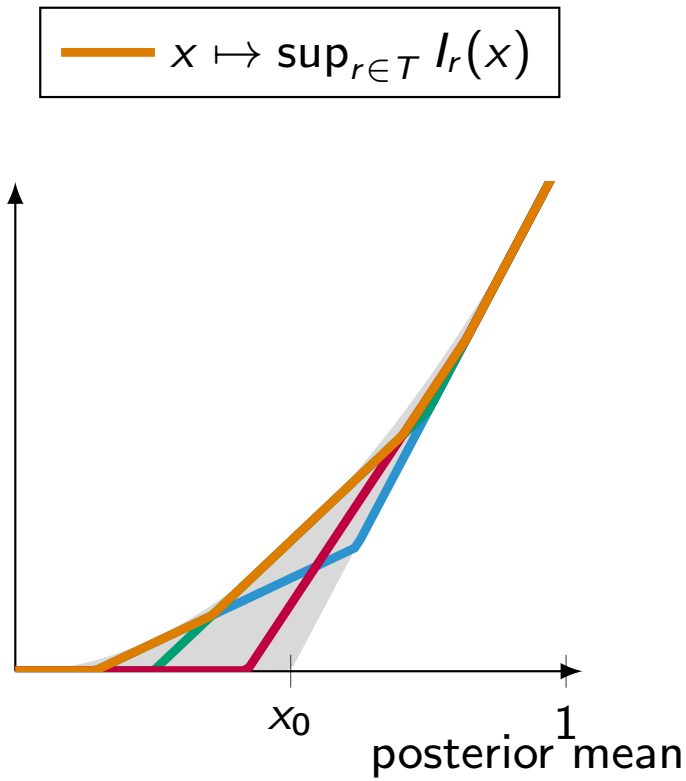
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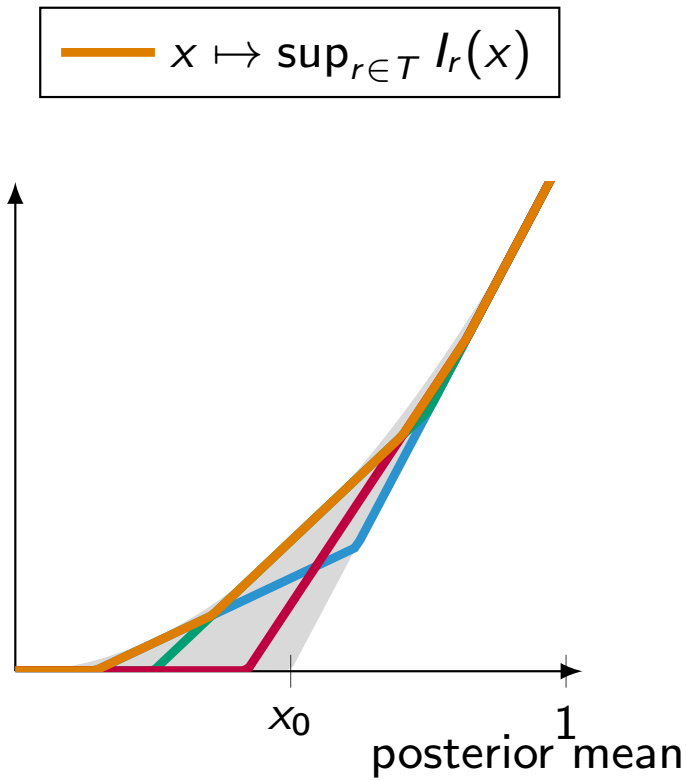
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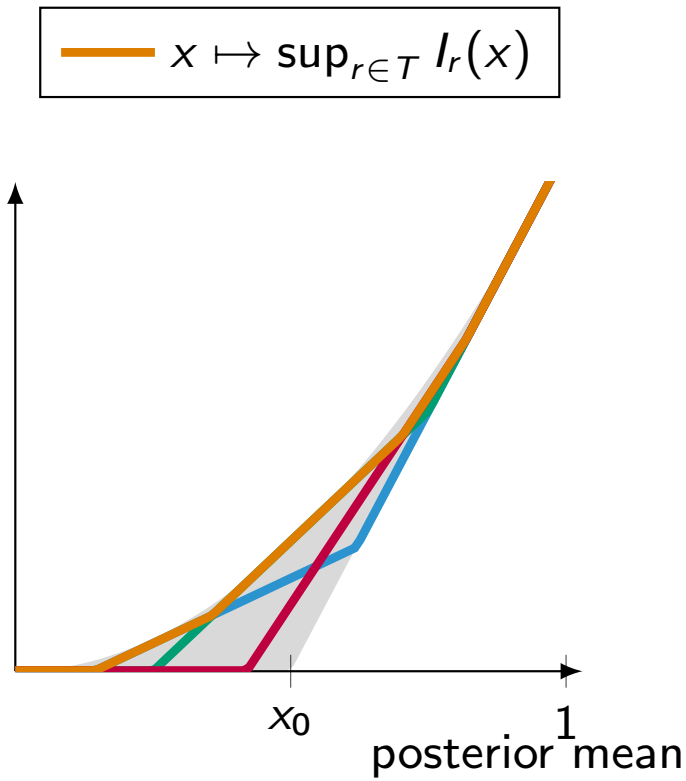
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If $\lambda = 0$, then: Kolotilin et al. (2017).

Optimal information policy

Sender's problem

The Sender's expected payoff from information policy I is

$$V_S(I) = \int_T \underbrace{e^*(I, c, \lambda)}_{\text{extensive margin}} \underbrace{\left(1 - I'_-(c) - \mathbf{1}\{x_0 \geq c\}\right)}_{\text{intensive margin}} + \mathbf{1}\{x_0 \geq c\} \, \mathrm{d}H(c, \lambda).$$

Assumption: (1) str. convex k; (2) Receiver chooses action 1 if indifferent.

I is optimal if I solves

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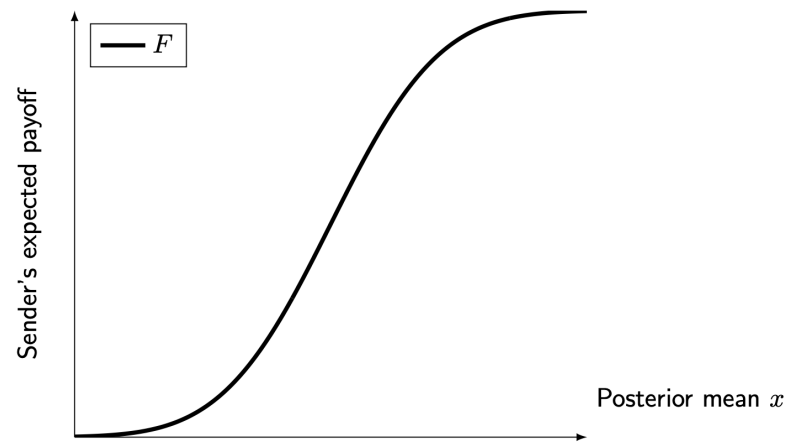
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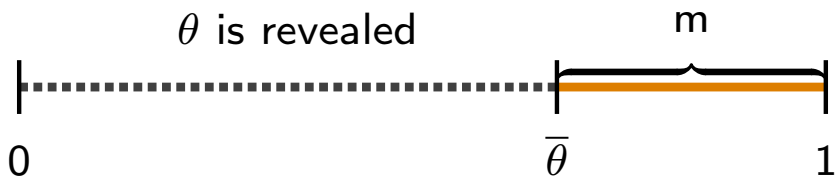
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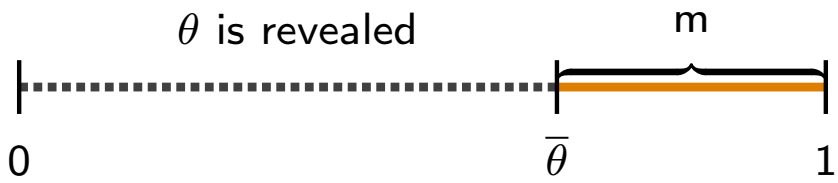
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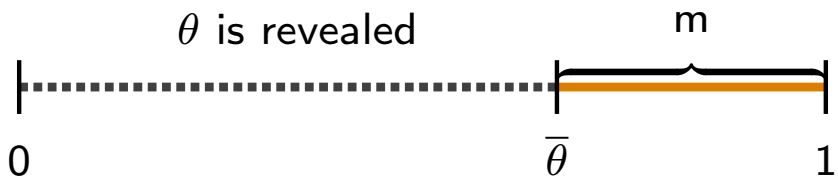
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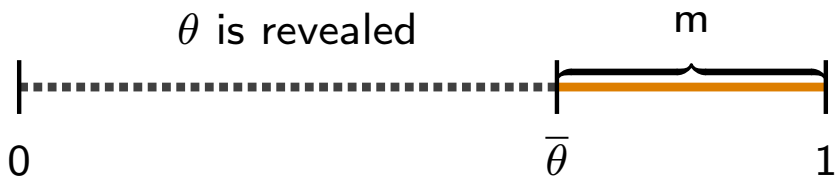
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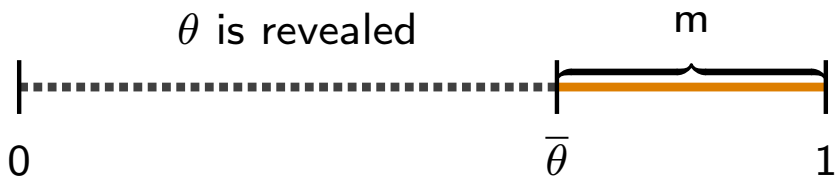
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↑ Not a linear function of I (Augias and Uhe, 2025).

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(2) k is linear.

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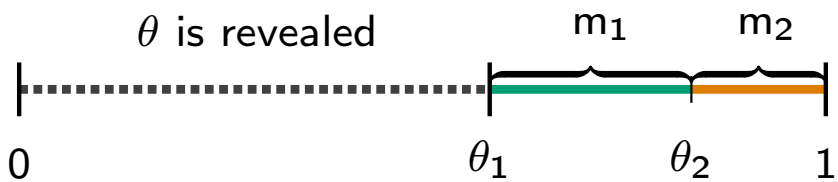
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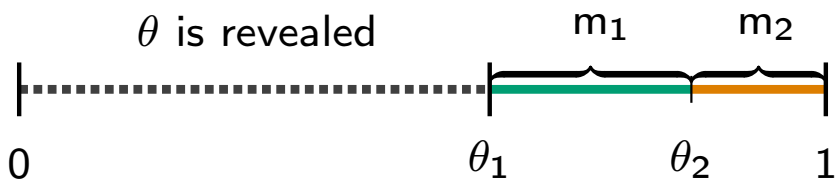
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Kolotilin et al. (2022): no media market;
Gehlbach and Sonin (2014): Sender knows c, and does not know λ.

Generalized principal-agent problem and coordination mechanisms

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- A coordination mechanism (R, M, π) is *direct* if: $R = T$, $M = [0, 1]$;
- A direct coordination mechanism π is IC if:
  1. truthful reporting;
  2. obeying the effort recommendation;

  are optimal for Receiver.

Proposition 4 (Myerson 1982.)

For every coordination mechanism G, for every equilibrium Sender's payoff v in G, there exists a direct and IC coordination mechanism π such that

$$\underbrace{V(\pi)}_{\substack{\text{Sender's payoff in} \\ \text{the truthful \& obedient eq'm of } \pi}} = v.$$

Deterministic coordination mechanisms

- A coordination mechanism (R, M, π) is *deterministic* if there exists $(I_r)_{r \in T}$ such that, for all r, $\pi(\{I_r\} \times M | r) = 1$.
- An IC persuasion mechanism $(I_r)_{r \in T}$ induces a Sender's payoff v if:

$$v = \int_T e^*(I_{(c,\lambda)}, c, \lambda)(1 - (I_{(c,\lambda)})'_-(c) - \mathbf{1}\{x_0 \geq c\}) + \mathbf{1}\{x_0 \geq c\} \, \mathrm{d}H(c, \lambda).$$

Proposition 5

For every deterministic coordination mechanism G, for every equilibrium Sender's payoff v in G, there exists a persuasion mechanism that induces the Sender's payoff v.

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Deterministic coordination mechanisms are equivalent to signals.

Correlating recommendation and information

Example: Linear k, uniform θ, $c = 3/4$ known to Sender, and two types:

Low ($\lambda_1 \approx 0.01$), Prob $= 8/9$;

High ($\lambda_2 = 0.02$), Prob $= 1/9$.

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There exists a **coordination mechanism** such that

Low gets a deterministic recommendation, 1, and some (low) information I_L.

High gets a **lottery**:

- Prob $p \approx 0.3$: Recommend $e = 1$ and the very informative J_H;
- Prob $1 - p \approx 0.7$: Recommend $e = 0$ and no information;

which improves upon any deterministic coordination mechanism.

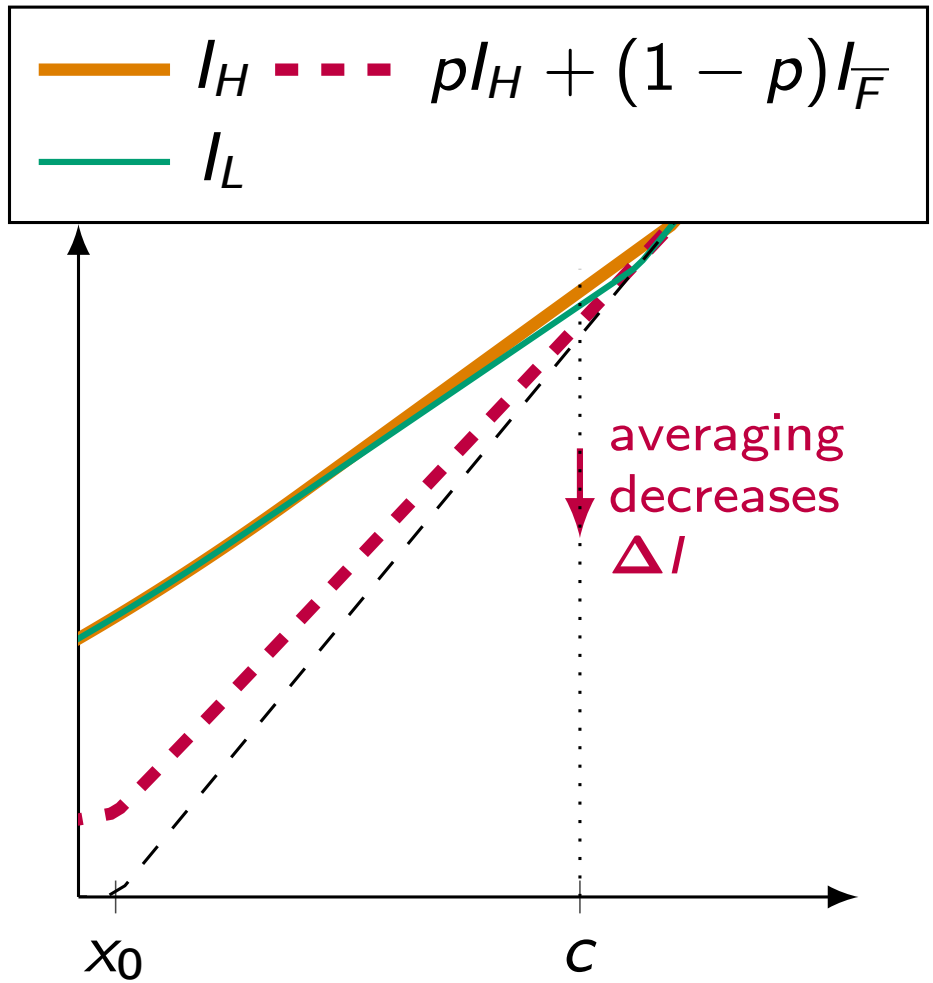
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Example: Linear k, uniform θ, $c = 3/4$ known to Sender, and two types:

Low ($\lambda_1 \approx 0.01$), Prob $= 8/9$;

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- High (weakly) prefers the lottery because ΔI_L does not cover λ;
- Low prefers I_L and effort 1 with probability 1 to the lottery;



Alternative approaches to inattention

Cost of learning about the state (Wei, 2021):

- Costly entropy reduction about the state;
- Flexible information acquisition about θ;

Interpretation: **Delegated learning** (e.g., analyzing a dataset).

Cost of processing the message (Bloedel and Segal, 2021):

- Costly entropy reduction about the message;
- Flexible information acquisition about m;

Interpretation: **Learning from communication** (e.g., decoding ads).

Cost of exposure:

- Cost $\lambda k(e)$ is independent of the informativeness;
- Receiver chooses mixtures of full and null observation;

Interpretation: **Learning via exposure** (e.g., scrolling feed, platform fee).

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Thanks!

Conclusion

- With inattention and private information:
  1. Persuasion mechanisms are equivalent to signals;
  2. Upper censorships are optimal;
  3. Persuasion mechanisms are without loss under deterministic coordination mechanisms;
- In the paper:
  1. Comparative statics wrt λ;
  2. Persuasion mechanisms are without loss if c is known and k is linear;
  3. "Experimental mechanisms": a subset of direct and IC stochastic coordination mechanisms that is without loss.

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Thanks!

Extra slides

Setting

Let $X := [0,1]$ be the state space, and F_0 a continuous prior CDF on X with mean $x_0 \in (0,1)$.

Definition 1 (Information Policies)

Let $\mathcal{C}(X)$ be the space of real-valued continuous functions on X endowed with the supremum norm $|\cdot|_\infty$. The uninformative policy is $\underline{I}(x) := (x - x_0)_+$, and the fully informative policy is $I_0(x) := \int_0^x F_0(t)\,\mathrm{d}t$. The set of feasible information policies is

$$\mathcal{I} := \{I \in \mathcal{C}(X) \mid I \text{ is convex},\ \underline{I} \leq I \leq I_0,\ \partial I(X) \subseteq [0,1]\}.$$

For any $I \in \mathcal{I}$, let I' denote its right-derivative.

Lemma 3 (Lemma 1)

The mapping $T\colon F \mapsto \left(x \mapsto \int_0^x F(y)\,\mathrm{d}y\right)$ is a homeomorphism from Bayes-plausible posterior-mean CDFs under the L^1 norm to $\langle \mathcal{I}, |\cdot|_\infty \rangle$.

Equilibrium

Let $T \coloneqq X \times \mathbb{R}_+$ be the type space endowed with a probability measure P. Given $I \in \mathcal{I}$ and effort $e \in [0,1]$, define the effective posterior-mean CDF $(e \odot I') \coloneqq eI' + (1-e)\overline{F}$, where $\overline{F}(x) \coloneqq \mathbf{1}\{x \geq x_0\}$.

Definition 2 (Equilibrium)

An equilibrium is a tuple $\langle I^\star, e, \alpha \rangle$, where $I^\star \in \mathcal{I}$, $e\colon T \times \mathcal{I} \to [0,1]$ is measurable, and $\alpha\colon T \times X \to [0,1]$ is measurable, such that:

1. **Action Optimality:** For all $x \in X$ and $(c,\lambda) \in T$,
$$\alpha(c,\lambda,x) \in \operatorname*{argmax}_{b \in [0,1]} b(x-c).$$
2. **Effort Optimality:** For all $(c,\lambda) \in T$ and $\hat{I} \in \mathcal{I}$,
$$e(c,\lambda,\hat{I}) \in \operatorname*{argmax}_{\tilde{e} \in [0,1]} \int_X (x-c)_+ \,\mathrm{d}(\tilde{e} \odot \hat{I}')(x) - \lambda k(\tilde{e}).$$
3. **Sender Rationality:** $I^\star \in \operatorname{argmax}_{\hat{I} \in \mathcal{I}} \hat{W}(\hat{I}, e, \alpha)$, where
$$\hat{W}(\hat{I}, e, \alpha) \coloneqq \int_T \int_X \alpha(c,\lambda,x) \,\mathrm{d}(e(c,\lambda,\hat{I}) \odot \hat{I}')(x) \,\mathrm{d}P(c,\lambda).$$

Receiver's Value and the Extensive Margin

Definition 3

The net informativeness operator is $\Delta\colon \mathcal{I} \to \mathcal{C}(X)$, defined by $\Delta I := I - \underline{I}$. The receiver's value function $V_\lambda : [0,1] \times \mathbb{R}_+ \to \mathbb{R}$ is:

$$V_\lambda(e, \Delta I(c)) := e\Delta I(c) - \lambda k(e).$$

Let $E_\lambda(\Delta I(c)) := \operatorname{argmax}_{e\in[0,1]} V_\lambda(e, \Delta I(c))$.

Lemma 4 (Lemmata 2 and 3)

The marginal benefit of effort $B(I,c) := \int_X (x-c)_+ \,\mathrm{d}I'(x) - (x_0 - c)_+$ satisfies $B(I,c) = \Delta I(c)$. Furthermore:

1. $V_\lambda(e, \Delta I(c))$ exhibits strictly increasing differences in $(e, \Delta I(c))$.
2. Every selection from the correspondence $E_\lambda(\cdot)$ is nondecreasing.
3. Given an equilibrium $\langle I^\star, e(\cdot), \alpha\rangle$ and $\lambda \geq 0$, the active set $\{c \in X \mid e(c, \lambda, I^\star) > 0\}$ is an interval, and contains x_0 if non-empty.

Optimality of Upper Censorships

Assumption 1 (SPness)

For all $\lambda \geq 0$, the conditional distribution of c given λ admits an absolutely continuous density $g(\cdot \mid \lambda)$. There exists a peak $p \in (0,1)$ such that, for all λ, $g(\cdot \mid \lambda)$ is nondecreasing on $[0,p]$ and nonincreasing on $[p,1]$.

Definition 4

For any state $\overline{\theta} \in X$, the $\overline{\theta}$-upper censorship $I_{\overline{\theta}} \in \mathcal{I}$ is the unique information policy satisfying:

$$I_{\overline{\theta}}(x) = \begin{cases} I_0(x), & \text{if } x \in [0,\overline{\theta}] \\ \max\Big\{ I_0(\overline{\theta}) + F_0(\overline{\theta})(x-\overline{\theta}),\ \underline{I}(x) \Big\}, & \text{if } x \in (\overline{\theta},1]. \end{cases}$$

Proposition 6 (Proposition 2)

Let Assumption 1 hold. There exists an equilibrium $\langle I^\star, e(\cdot), \alpha \rangle$ wherein $I^\star$ is a $\overline{\theta}$-upper censorship. Moreover, for any two equilibria $\langle I_1^\star, e_1, \alpha_1 \rangle$ and $\langle I_2^\star, e_2, \alpha_2 \rangle$, $\hat{W}(I_1^\star, e_1, \alpha_1) = \hat{W}(I_2^\star, e_2, \alpha_2)$.

Application

Setup: Sender's utility is $U_S(a, e) = a + \gamma e$ for $\gamma \geq 0$. Cost is linear: $k(e) = e$. Let $V_S(I, e, \alpha)$ denote the Sender's ex-ante expected utility under U_S.

Definition 5

An information policy $I \in \mathcal{I}$ is a bi-upper censorship if there exist $0 \leq \theta_1 \leq x_1 \leq x_2 \leq 1$ and $m = \frac{\underline{I}(x_2) - [I_0(\theta_1) + F_0(\theta_1)(x_1 - \theta_1)]}{x_2 - x_1}$ such that:

$$I(x) = \begin{cases} I_0(x), & \text{if } x \in [0, \theta_1] \\ I_0(\theta_1) + F_0(\theta_1)(x - \theta_1), & \text{if } x \in (\theta_1, x_1] \\ \underline{I}(x_2) - m(x_2 - x), & \text{if } x \in (x_1, x_2] \\ \underline{I}(x), & \text{if } x \in (x_2, 1]. \end{cases}$$

Proposition 7 (Proposition 3)

Let $P(X \times \{\lambda^\star\}) = 1$ for some $\lambda^\star \geq 0$. Let Assumption 1 hold with $p \geq x_0$. For every equilibrium $\langle I^\star, e(\cdot), \alpha \rangle$, there exists a bi-upper censorship $\hat{I} \in \mathcal{I}$ such that $V_S(\hat{I}, e, \alpha) \geq V_S(I^\star, e, \alpha)$.

Coordination mechanisms

Definition 6

A coordination mechanism is a tuple $G = (R, \mathcal{R}, M, \mathcal{M}, \pi)$ where $(R, \mathcal{R})$ is a measurable space of reports, $(M, \mathcal{M})$ is a measurable space of messages, and $\pi\colon R \to \Delta(\mathcal{I} \times M)$ is a Markov kernel. G is *direct* if $(R, \mathcal{R}) = (T, \mathcal{B}_T)$ and $(M, \mathcal{M}) = ([0,1], \mathcal{B}_{[0,1]})$.

Definition 7

A participation strategy (ρ, δ) consists of measurable functions $\rho\colon T \to R$ and $\delta\colon T \times M \to [0,1]$. It is an equilibrium in G if no type $t \in T$ prefers any measurable deviation $\eta\colon M \to [0,1]$ and a report $r \in R$. A direct mechanism is Incentive Compatible (IC) if truth-telling and obedience $(\rho^\star(t) = t, \delta^\star(t,m) = m)$ forms an equilibrium.

Proposition 8 (Proposition 4)

For every coordination mechanism G and every equilibrium (ρ, δ) in G, there exists an IC direct coordination mechanism $F = (T, [0,1], \pi^F)$ such that $V_0^F(\rho^\star, \delta^\star) = V_0^G(\rho, \delta)$, in which V_0 denotes the Sender's ex-ante expected utility.

Deterministic mechanisms

A *product coordination mechanism* is a coordination mechanism (R, M, π) such that, for all $r \in R$, the probability measure $\pi(\cdot|r)$ is a product measure.

Proposition 9 (Proposition 5)

For every product coordination mechanism G, for every equilibrium (ρ, δ) in G, there exists an IC persuasion mechanism $I_\bullet$ and a payoff $v \in \mathbb{R}_+$ such that: $I_\bullet$ induces v and $v \geq V_0^G(\rho, \delta)$.

Experimental and activation mechanisms

A direct coordination mechanism $G = (T, E, \pi)$ is an *experimental mechanism* if: for all t, there exist a probability measure μ_t on $\mathcal{I}$ and a measurable function $f_t : \mathcal{I} \to E$ such that, for all $\overline{I} \in \mathcal{B}_{\mathcal{I}}$, $\overline{E} \in \mathcal{B}_E$, we have

$$\pi(\overline{I} \times \overline{E}|t) = \mu_t(f_t^{-1}(\overline{E}) \cap \overline{I}).$$

Proposition 10 (Experimental mechanisms are without loss)

For every direct incentive compatible coordination mechanism G, there exists an incentive compatible experimental mechanism F such that

$$V_0^F(\rho^*, \delta^*) = V_0^G(\rho^*, \delta^*).$$

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A direct coordination mechanism $G = (T, E, \pi)$ is an *activation mechanism* if: for all t, there exist a probability measure μ_t on E and a measurable function $f_t \colon E \to \mathcal{I}$ such that, for all $\overline{I} \in \mathcal{B}_{\mathcal{I}}$, $\overline{E} \in \mathcal{B}_E$, we have

$$\pi(\overline{I} \times \overline{E}|t) = \mu_t(f_t^{-1}(\overline{I}) \cap \overline{E}).$$

Proposition 11

For every direct incentive compatible coordination mechanism G, there exists an incentive compatible activation mechanism F such that

$$V_0^F(\rho^*, \delta^*) = V_0^G(\rho^*, \delta^*).$$

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